

1. Understanding Logistic Regression Model

Instead of predicting y directly, Logistic Regression models the probability:

$$P(y = 1|x) = h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

where:

- $h_{\theta}(x)$ is the **sigmoid function**, giving the probability of class 1.
- θ are the parameters (weights).
- x is the input feature vector.

For class 0, the probability is:

$$P(y = 0|x) = 1 - h_{\theta}(x)$$

2. Loss Function for a Single Example

We use **log-likelihood** to define the loss.

For **one data point** (x, y) , the likelihood of correct classification is:

$$L(\theta) = \begin{cases} h_{\theta}(x), & \text{if } y = 1 \\ 1 - h_{\theta}(x), & \text{if } y = 0 \end{cases}$$

Taking the **log** (to make it easier to optimize):

$$\log L(\theta) = \begin{cases} \log h_{\theta}(x), & \text{if } y = 1 \\ \log(1 - h_{\theta}(x)), & \text{if } y = 0 \end{cases}$$

We combine both cases into one equation:

$$\text{Loss} = -[y \log h_{\theta}(x) + (1 - y) \log(1 - h_{\theta}(x))]$$

This is called the **Binary Cross-Entropy Loss** (or Log Loss).

3. Cost Function for the Whole Dataset

For **multiple examples** (m samples), we take the **average loss** over all examples:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)}))]$$

where:

- m = number of training examples.
- $y^{(i)}$ = actual label (0 or 1) for the i -th sample.
- $h_{\theta}(x^{(i)})$ = predicted probability.

This function ensures that:

- If $h_{\theta}(x)$ is close to y (correct prediction) → Low cost.
- If $h_{\theta}(x)$ is far from y (wrong prediction) → High cost.

4. Why Use Logarithms?

- Avoids computational issues with probability values.
- Converts **multiplication into addition** (useful for optimization).
- Ensures **convexity** → allows for efficient optimization with Gradient Descent.

Final Summary

- **Loss Function** (for one example):

$$L = -[y \log h_{\theta}(x) + (1 - y) \log(1 - h_{\theta}(x))]$$

- **Cost Function** (for all examples):

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)}))]$$

This is optimized using **Gradient Descent** to find the best parameters θ .